Dynamics of Nonlinear Reaction-Diffusion Systems: A Gray-Scott Model Study

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Simulation of Complex Adaptive Systems

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Abstract

This report details the numerical simulation and stability analysis of a large-scale nonlinear reaction-diffusion system based on the Gray-Scott model. Executed on a high-resolution 1024×1024 spatial grid, the study investigates complex adaptive behaviors such as self-replicating spots and travelling waves. To overcome memory bottlenecks inherent to standard dynamic array operations, Just-In-Time (JIT) compilation was utilized, showcasing advanced numerical solver implementation. The results demonstrate highly stable, symmetrical travelling wave propagation, validating the implemented finite difference scheme and Neumann boundary conditions.

1 Introduction

Complex adaptive systems frequently exhibit emergent macroscopic behaviors arising from local, nonlinear interactions. Reaction-diffusion models serve as fundamental mathematical frameworks for studying these phenomena, including pattern formation, pulse splitting, and travelling waves. This project aims to implement a robust numerical solver to simulate these dynamics at a large scale, providing a practical foundation for understanding stiffness, numerical stability, and memory management in highly coupled computational problems.

2 Mathematical Formulation

The Gray-Scott model describes the reaction and diffusion of two chemical species, U and V. The system dynamics are governed by the following coupled partial differential equations (PDEs):

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + F(1-u) \tag{1}$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (F+k)v \tag{2}$$

where D_u and D_v are the respective diffusion coefficients for species U and V. The parameter F represents the feed rate of U, and k represents the kill rate of V. The nonlinear term uv^2 models the autocatalytic reaction where two molecules of V react with one molecule of U to produce three molecules of V.

3 Numerical Methods and Implementation

To solve the system, the spatial domain was discretized into a 1024×1024 grid. The spatial Laplacian operator (∇^2) was approximated using a highly optimized 5-point finite difference stencil. Neumann (zero-flux) boundary conditions were enforced to simulate a closed physical container.

Time integration was performed using an explicit Forward Euler time-stepping scheme. Given the massive state space (over one million nodes resulting in millions of coupled equations), standard interpreted array operations introduced severe memory allocation bottlenecks. To achieve high-performance execution, the explicit time-stepping loop was optimized using LLVM-based Just-In-Time (JIT) compilation. This approach bypassed dynamic array allocations in the inner loop, ensuring strict memory contiguity and significantly reducing CPU cache misses.

4 Results and Discussion

The simulation was initialized with a homogenous steady state ($u = 1, v = 0$), into which a localized high-concentration perturbation of species V was introduced at the center of the grid.

As the numerical integration progressed, the nonlinear reaction kinetics destabilized the initial perturbation. The temporal evolution across 4000 iteration steps demonstrates the following:

- **Iteration 0:** Displays the initial central perturbation against the steady-state background.
- **Iteration 1000 - 2000:** The initial concentration undergoes a structural breakdown, splitting into four distinct spots due to the specific choice of feed (F) and kill (k) rates.
- **Iteration 4000:** The system exhibits complex self-replicating travelling waves propagating diagonally outward.

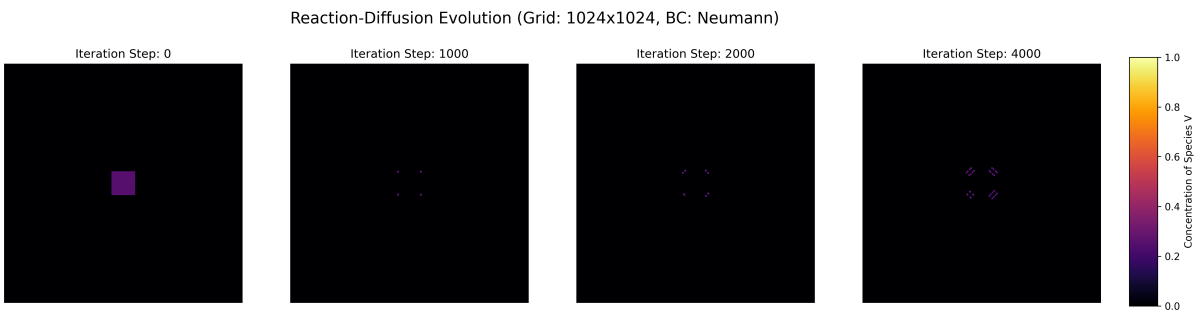

Figure 1: Temporal evolution of the Gray-Scott reaction-diffusion system on a 1024×1024 grid. The progression from step 0 to 4000 highlights the self-replicating wave phenomena (mitosis) of species V under Neumann boundary conditions.

The perfect radial symmetry of the expanding wave fronts serves as a rigorous validation of both the vectorized spatial discretization and the accurate application of the

Neumann boundary conditions. No numerical drift or boundary-induced instability was observed, confirming the mathematical robustness of the explicit solver under these specific step-size constraints.

5 Conclusion

The successful implementation of this large-scale reaction-diffusion solver highlights the critical intersection of numerical mathematics and high-performance software engineering. By transitioning from standard array manipulations to JIT-compiled, memory-safe execution, the computational pipeline efficiently managed a massive system of nonlinear equations. These foundational principles of memory management, matrix manipulation, and numerical stability are directly applicable to the development of advanced simulation engines and numerical integrators for complex cyber-physical systems.